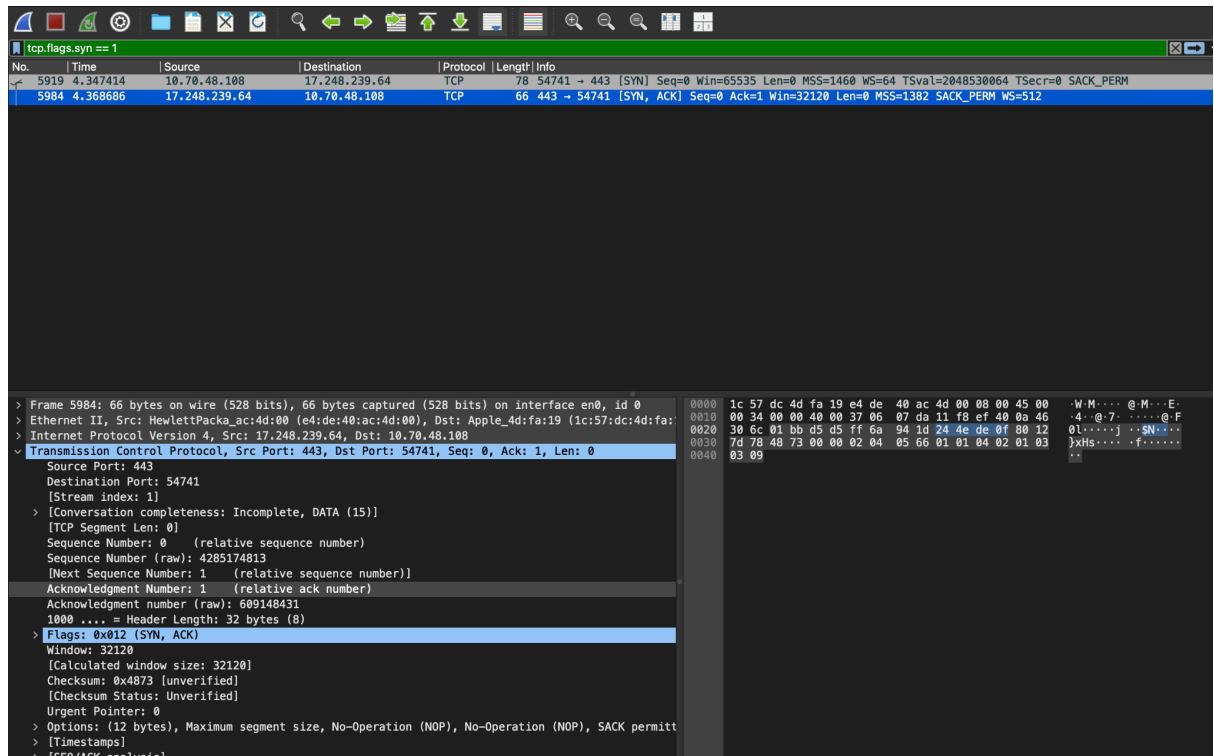


- Source: 10.70.48.108
Port Number of Source: 54740
- Destination: 128.119.245.12
Port Number of Destination: 80

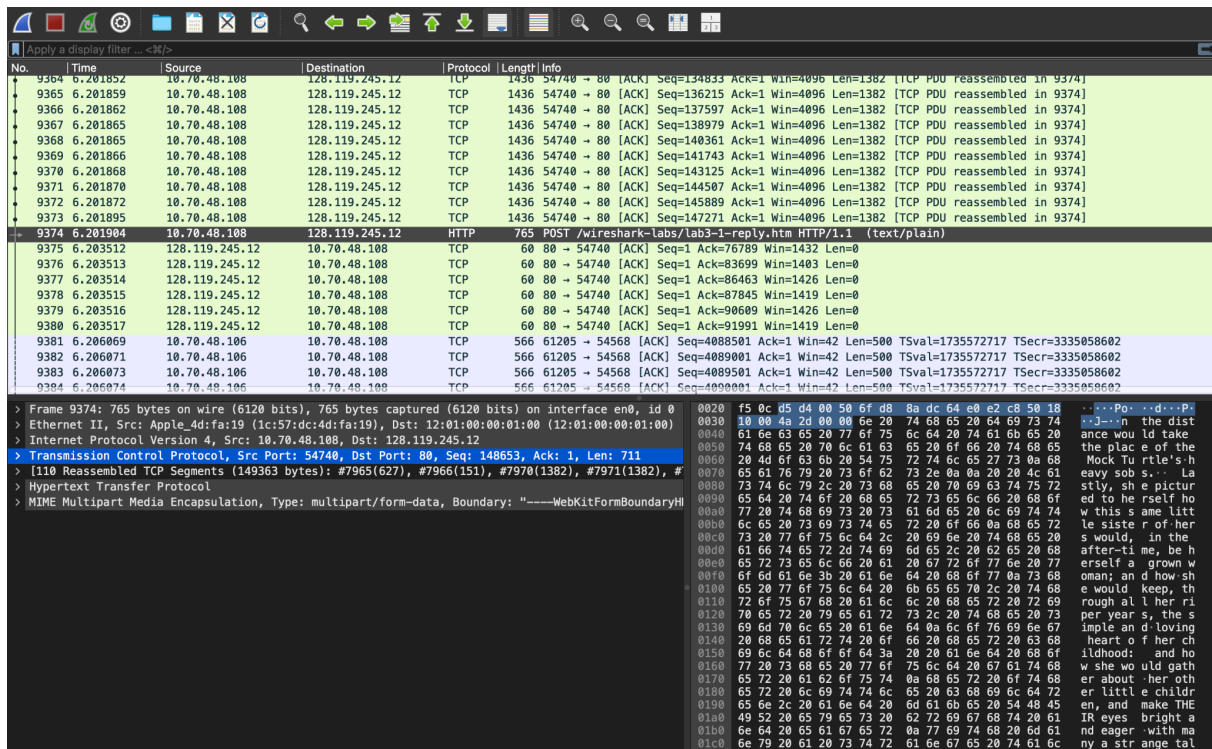
The screenshot shows a Wireshark packet capture. The packet list on the left shows a series of TCP ACKs from 10.70.48.108 to 128.119.245.12, followed by an HTTP 200 response. The packet details pane for packet 9374 shows the HTTP response structure, including the status line '200 OK' and the content type 'text/plain'.

- Source: 10.70.48.108
Port Number of Source: 54740
- Sequence Number = 0
Flag's value as 0x002 identifies the segment as a SYN segment.
- Sequence Number = 0
Acknowledgement Number = 1
Flag's value as 0x012 identifies the segment as a SYN ACK segment.

The screenshot shows a Wireshark packet capture with a display filter of 'tcp.flags.syn == 1'. The packet list shows a SYN packet (Seq=0) and a SYN-ACK packet (Seq=0, Ack=1). The packet details pane for the SYN-ACK packet shows the sequence number 0, acknowledgment number 1, and the SYN flag set.



6. Sequence Number for all 6 TCP segments is 1.



7. Sequence Number for all 6 TCP segments is 1.

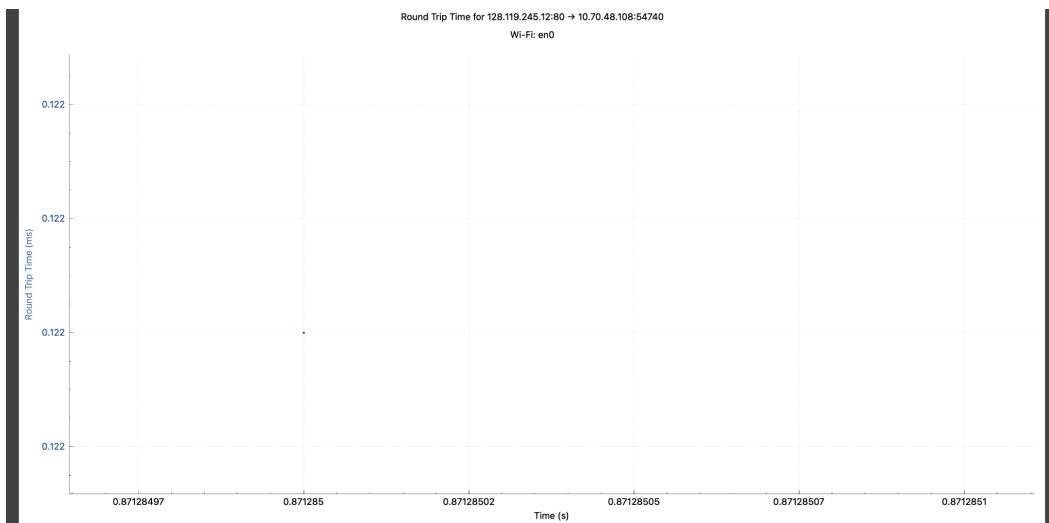
Time at which each segment was sent is:

- 1st segment: Feb 18, 2025 11:19:42.913608000 IST

- 2nd segment: Feb 18, 2025 11:19:42.913609000 IST
- 3rd segment: Feb 18, 2025 11:19:42.913610000 IST
- 4th segment: Feb 18, 2025 11:19:42.913611000 IST
- 5th segment: Feb 18, 2025 11:19:42.913612000 IST
- 6th segment: Feb 18, 2025 11:19:42.913613000 IST

RTT for each segment is as follows:

- 1st segment: 0.216722000 seconds
- 2nd segment: 0.216711000 seconds
- 3rd segment: 0.216708000 seconds
- 4th segment: 0.216708000 seconds
- 5th segment: 0.216703000 seconds
- 6th segment: 0.216651000 seconds



8. Length of each TCP Segment is 0
9. Minimum Calculated Window Size: 1403

```
Window: 1403
[Calculated window size: 1403]
[Window size scaling factor: -1 (unknown)]
```

10. Yes. There are retransmitted segments in the trace file. It was checked by "tcp.analysis.retransmission".
11. 6910 bits of data does the receiver typically acknowledge in an ACK
12. Throughput is calculated as: 152861 bytes / 16.21 seconds = 9444.68 bytes per seconds. It was calculated as bits length between first packet and last packet and the same was done with the time calculation.

13.

